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Fossil Fuel Prices and Capital Cost

A Machine Learning Driven Study on Energy Transition

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Research question

How do fossil fuel prices relate to energy-project financing costs?

- Debt and equity providers require a return on the capital they commit.
- A higher financing cost raises the revenue a project must earn.
- Regional financing differences can change modeled transition pathways.

WACC

Weighted average cost of capital

$$\underbrace{Lr_D(1-\tau)}_{\text{debt}} + \underbrace{(1-L)r_E}_{\text{equity}}$$

Ameli et al., Nature Communications (2021); Calcaterra et al., Nature Energy (2024).

Price transmission channels

What links fossil fuel prices to energy-project WACC?



We compare these signals across China, Europe, and the Middle East.

Mechanisms from Melodia and Karlsson (2022), Lieberman and Doherty (2007), and Duma and Muñoz Cabré (2023).

Data and scope

Input	Content	Role
Fuel prices	Monthly coal, oil, and natural gas series	Price-forecast inputs
Macroeconomy	Inflation, growth, unemployment, interest-rate and monetary indicators	Regional inflation forecasts
WACC	Calcaterra et al. model projections, aligned to 2026–2035	Financing scenarios

3 regions China, Europe, Middle East

120 monthly rows per cleaned region

9 technologies wind, solar, hydro, biomass, nuclear, CCS, coal, and gas

WACC scenario data: Calcaterra et al. (2024). Coal and gas generation provide conventional-technology comparisons.

Three-stage framework

1 Fuel prices

Sequence model
CNN + LSTM/GRU
Attention + GPR
smoothing

2 Inflation

Regional LSTM
Feed-forward predictor
Five-model ensemble

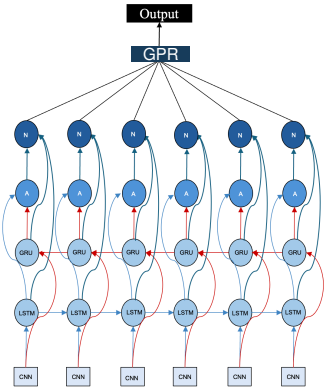
3 WACC

Fuel price + inflation
AutoGluon regression
Region–technology
comparisons

Comparison structure 3 regions × 3 fuels × 9 technologies = 81 combinations

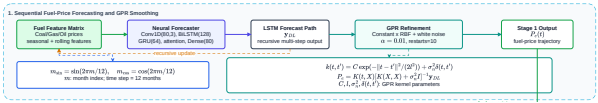
Paper Figs. 1–2.

Fuel-price forecasting model



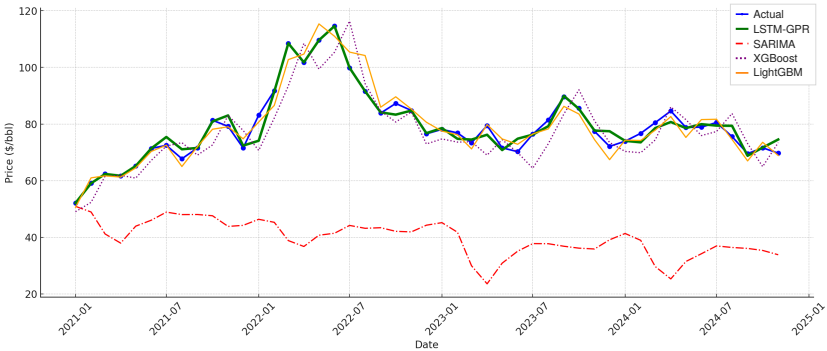
Temporal learning and GPR smoothing

- A 12-month input window
- CNN, recurrent layers, and attention
- Seasonal and rolling-price features
- Gaussian-process smoothing



Paper Figs. 2–3.

Oil-price comparison, 2021–2024



0.963
R^2

1.86%
MAPE

2.416
RMSE (USD/bbl)

Paper Fig. 4; metrics recomputed from the 47 saved monthly observations.

WACC estimation and validation

AutoGluon regression

$$\mathrm{WACC}_{r,p,t} = f_{r,p,e}(P_{e,t}, \pi_{r,t})$$

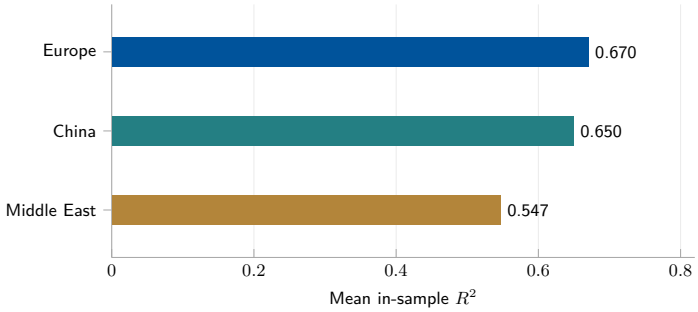
- One fuel price and an inflation covariate
- Bagging and stacking
- In-sample R^2, MAPE, and RMSE

Time-split validation

- OLS/Ridge with cleaned regional inputs
- Train 2026–2032; test 2033–2035
- Rolling one-year-ahead checks
- Compare fuel + inflation + trend against a time trend

Original AutoGluon fit and cleaned-input OLS/Ridge validation are reported separately.

Regional results



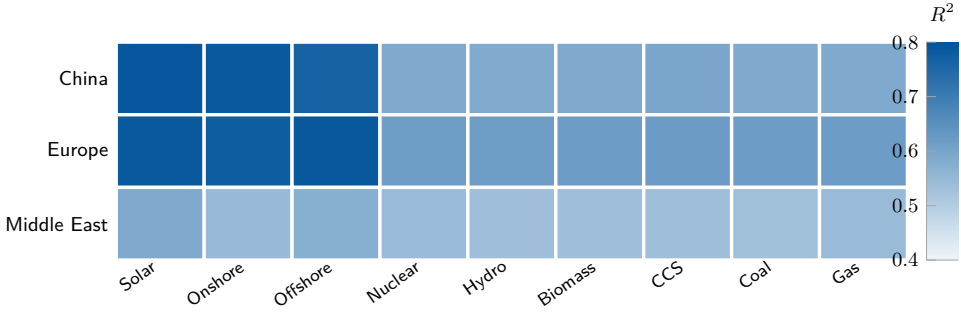
Europe and China show stronger fit.

Middle Eastern specifications have a lower average fit and larger percentage errors.

Mean R^2 across the 27 fuel–technology specifications per region.

Paper Fig. 5 and results workbook; joint fuel-price and inflation inputs. R^2 measures in-sample model fit.

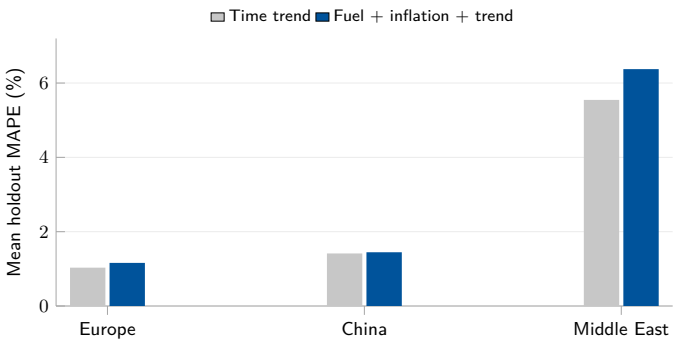
Technology results



Solar 0.718 **Offshore wind** 0.706 **Onshore wind** 0.700

Technology means across three fuels and three regions; heatmap cells average across fuels. Nuclear, hydro, biomass, CCS, coal, and gas are around 0.58 overall.

Validation over time



Improvement over a time trend

Positive incremental out-of-sample R^2:

China	8/27
Europe	6/27
Middle East	6/27

Training: 2026–2032.
Testing: 2033–2035.

Ridge diagnostics on cleaned scenario panels; fixed holdout: 2033–2035. Counts compare fuel + inflation + trend against time trend alone.

Macroeconomic transmission

Target	Fuel	Lag	p-value
China real interest rate	Coal	1 year	0.00018
China real interest rate	Coal	2 years	0.00691
China real interest rate	Gas	1 year	0.01915
Europe inflation	Gas	2 years	0.00952

Selected links appear through inflation and interest rates.

Selected Granger-style diagnostics on annual differenced targets and log-differenced fuel prices. Unadjusted exploratory p-values.

Regional policy implications

Region	Relevant context	Financing question
Europe	Imported energy and policy credibility	How can predictable contracts and risk transfer support investment?
China	International spillovers and domestic price regulation	How do public finance and price rules alter exposure?
Middle East	Oil dependence and domestic energy-pricing institutions	How can financing capacity support diversification?

Polzin et al. (2019); He and Lin (2017); Karanfil and Omgba (2023); Fattouh and El-Katiri (2013).

Selected references

1. N. Ameli et al., “Higher cost of finance exacerbates a climate investment trap in developing economies,” *Nature Communications*, 12, 4046, 2021. doi:10.1038/s41467-021-24305-3
2. M. Calcaterra et al., “Reducing the cost of capital to finance the energy transition in developing countries,” *Nature Energy*, 9, 1241–1251, 2024. doi:10.1038/s41560-024-01606-7
3. L. Paranhos, “Predicting inflation with recurrent neural networks,” *International Journal of Forecasting*, online 2025. doi:10.1016/j.ijforecast.2024.07.010
4. F. Polzin et al., “How do policies mobilize private finance for renewable energy?” *Applied Energy*, 236, 1249–1268, 2019. doi:10.1016/j.apenergy.2018.11.098

Conclusions

1. Fuel-price and inflation forecasts feed regional WACC scenarios.
2. Fit differs across regions and technologies.
3. Time-split checks show where price signals add information.

Code + materials

Dataset snapshot

Paper, code, data, and editable LaTeX sources.

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